

Global CO₂ Emissions

Who Emits, How Much, and What's Changing

Scope: Fossil fuels & industry · 1750–2024

Data: Global Carbon Budget 2025 · Our World in Data

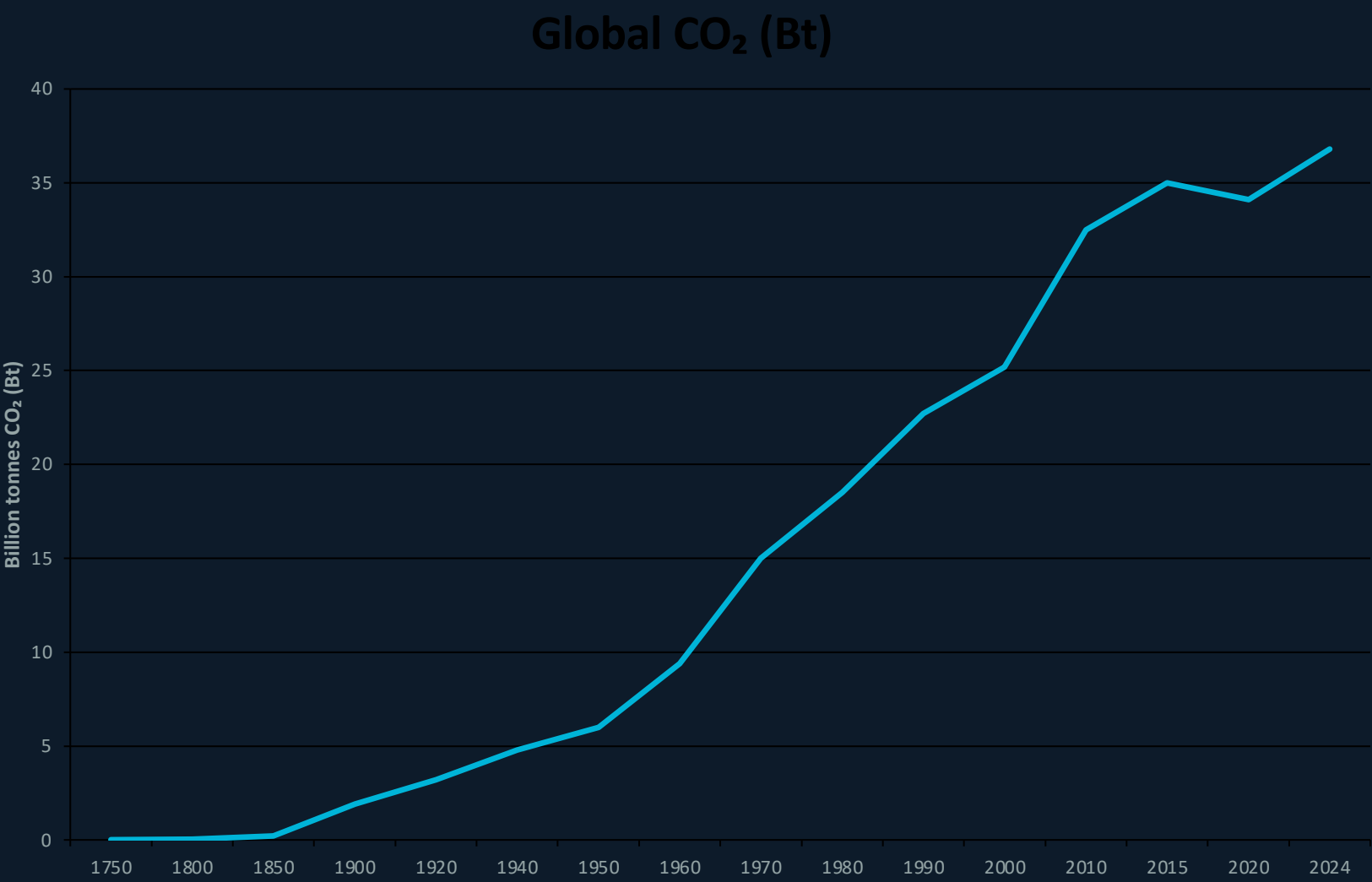
For: Policy analysts · Climate negotiators · Sustainability researchers

>35 Bt CO₂

emitted annually in 2024

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Fossil fuels & industry · 1750–2024 | Global Carbon Budget 2025



1950

6 Bt

Post-war industrialisation begins

1990

22+ Bt

Near 4× growth in 40 years

2024

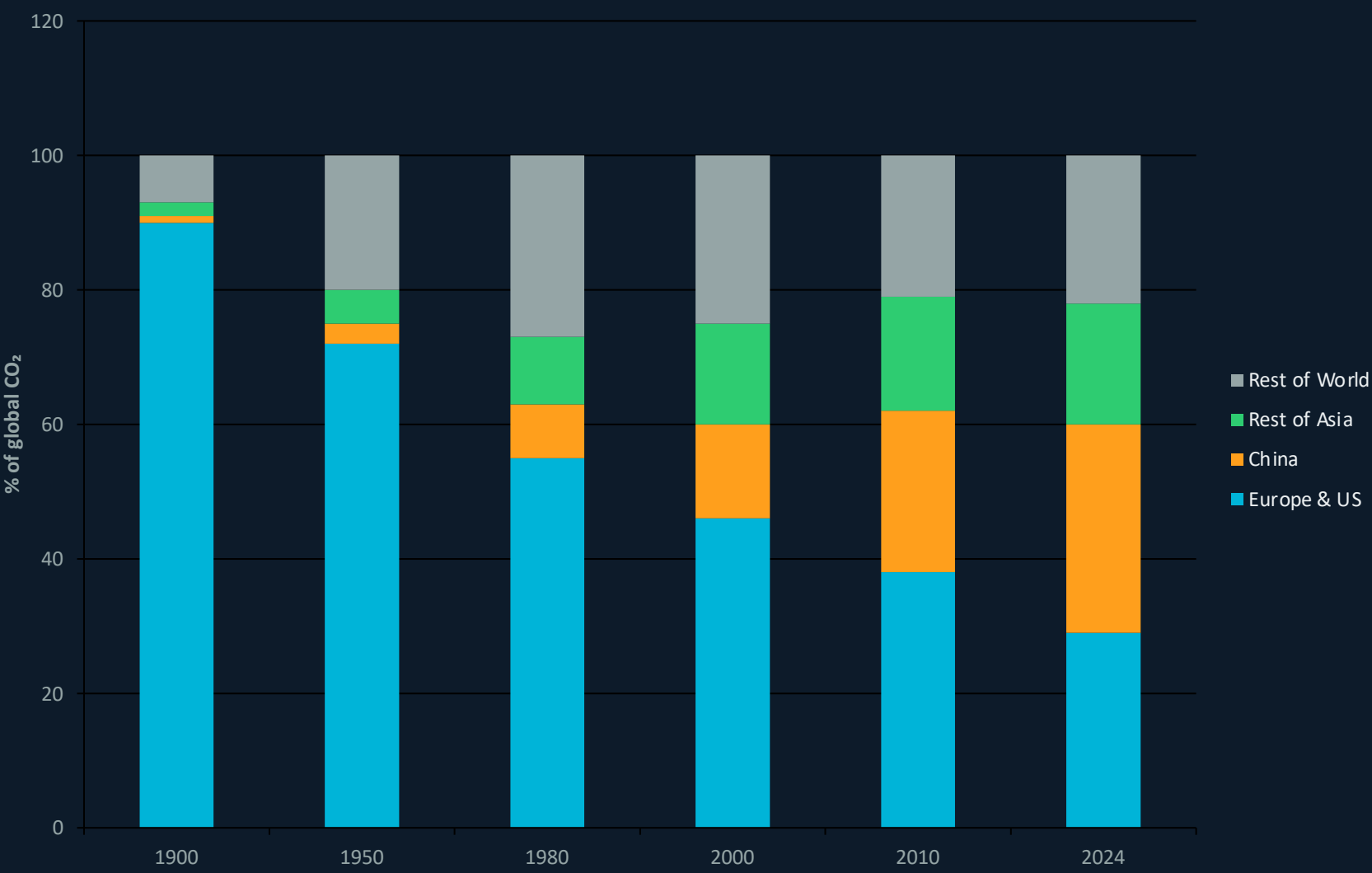
35+ Bt

No peak yet; growth slowing

Land-use change emissions declining slightly; fossil fuels dominate the total.

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

Share of global CO₂ emissions by region | Global Carbon Budget 2025



Europe & US

90% → <29%

Industrialised early; now declining share

China

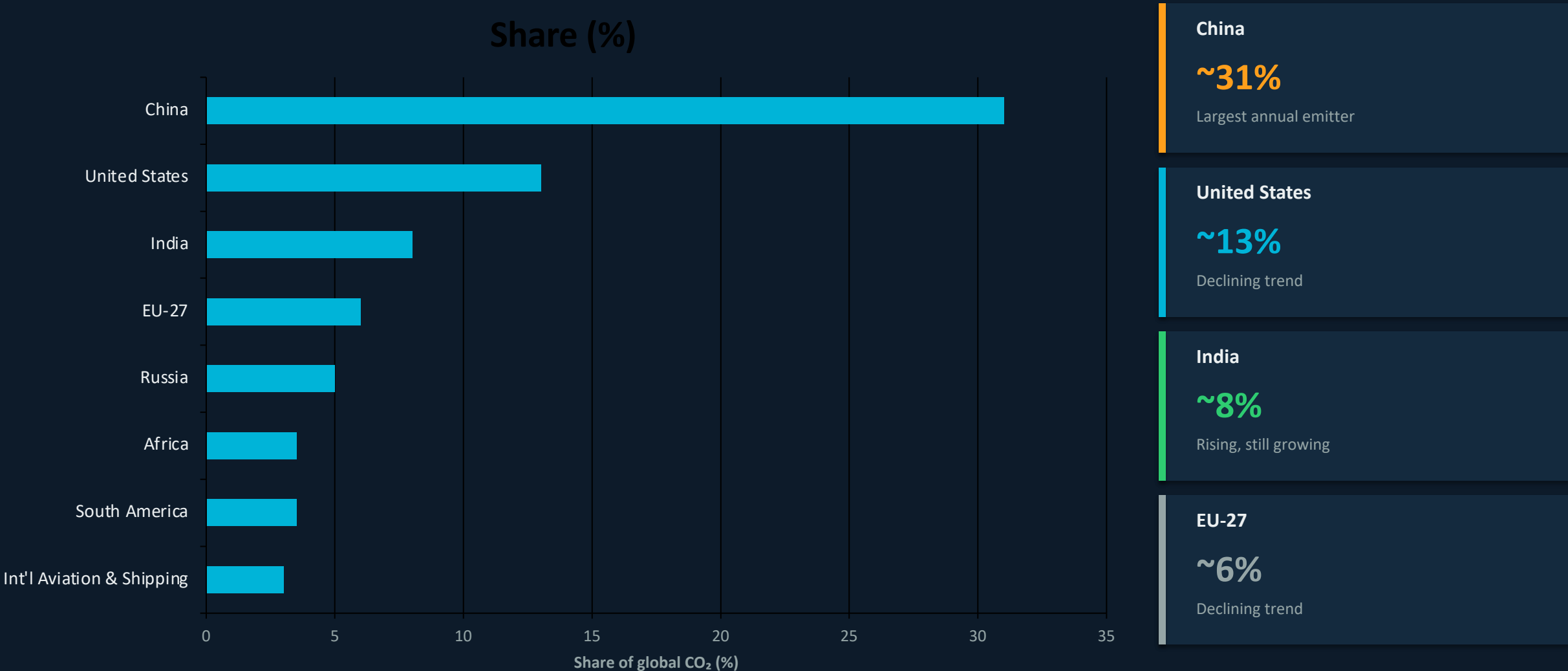
~1% → 31%

World's largest annual emitter since ~2006

Africa and South America each account for only 3–4% of global emissions.

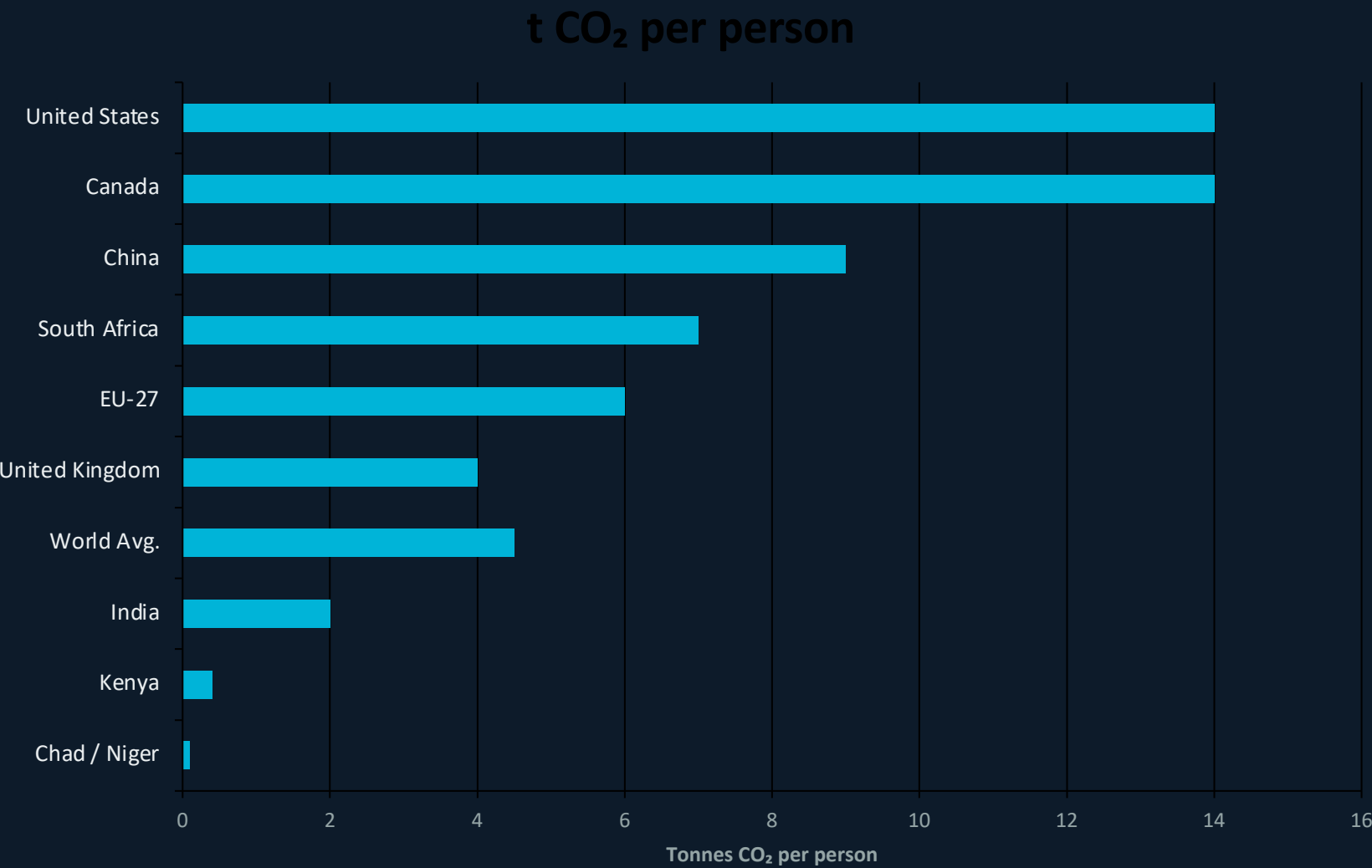
China Now Emits More Than One-Quarter of Global CO₂

Country & regional shares of global CO₂ emissions, 2024 | Global Carbon Budget 2025



Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

Per capita CO₂ emissions, selected countries (approx. 2024) | Global Carbon Budget 2025



150×

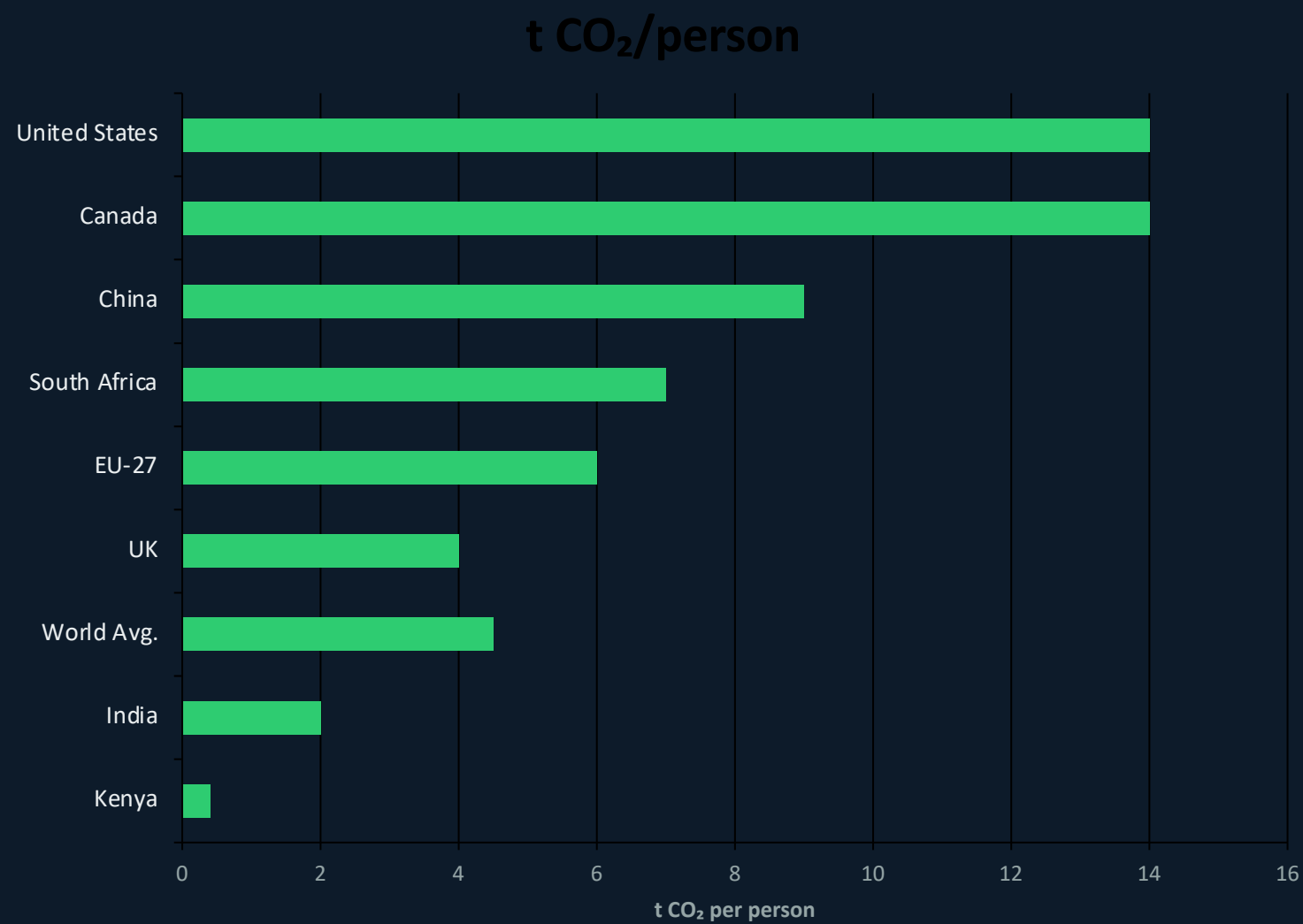
gap between Chad / Niger
(~0.1 t) and US / Canada (~14 t)

Similar prosperity,
different emissions

France, UK & Portugal emit far less per capita than Germany, Netherlands or the US — due to more nuclear & renewables.

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

Per capita CO₂ snapshot (approx. 2024) | Global Carbon Budget 2025



US / Canada

~14 t/person

3× world average

China

~9 t/person

Below US despite being #1 total emitter

India

~2 t/person

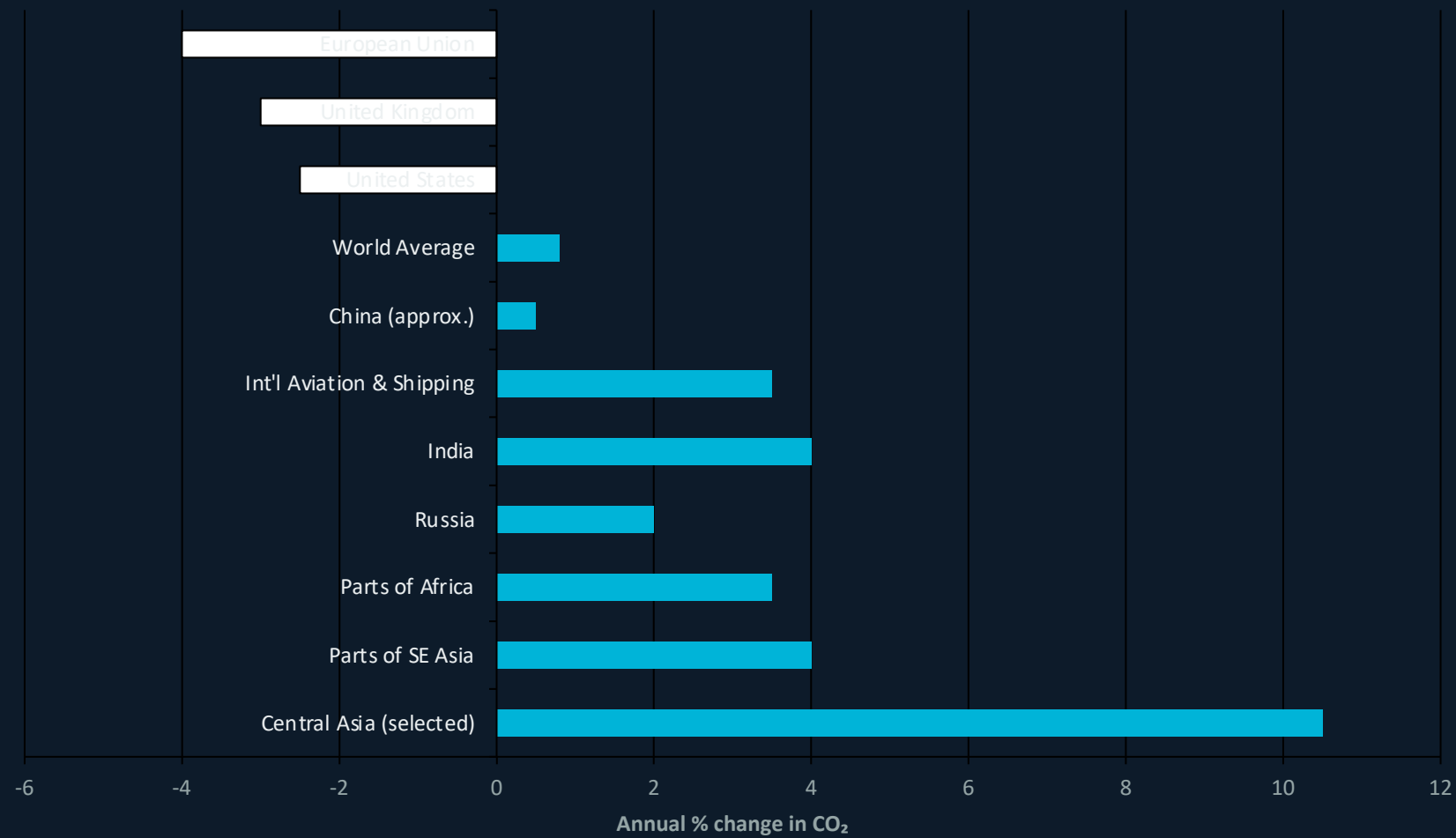
Well below world avg. of 4.5 t

Source: figure_8.jpg (Our World in Data) shows how per capita emissions evolved since 1750.

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

Annual % change in CO₂ emissions, 2024 | Global Carbon Budget 2025

% change



2024 Trends

- Increases (red/orange):**
Russia, SE Asia, Central Asia, parts of Africa
- Declines (blue):**
US, EU, UK — policy & renewable energy gains

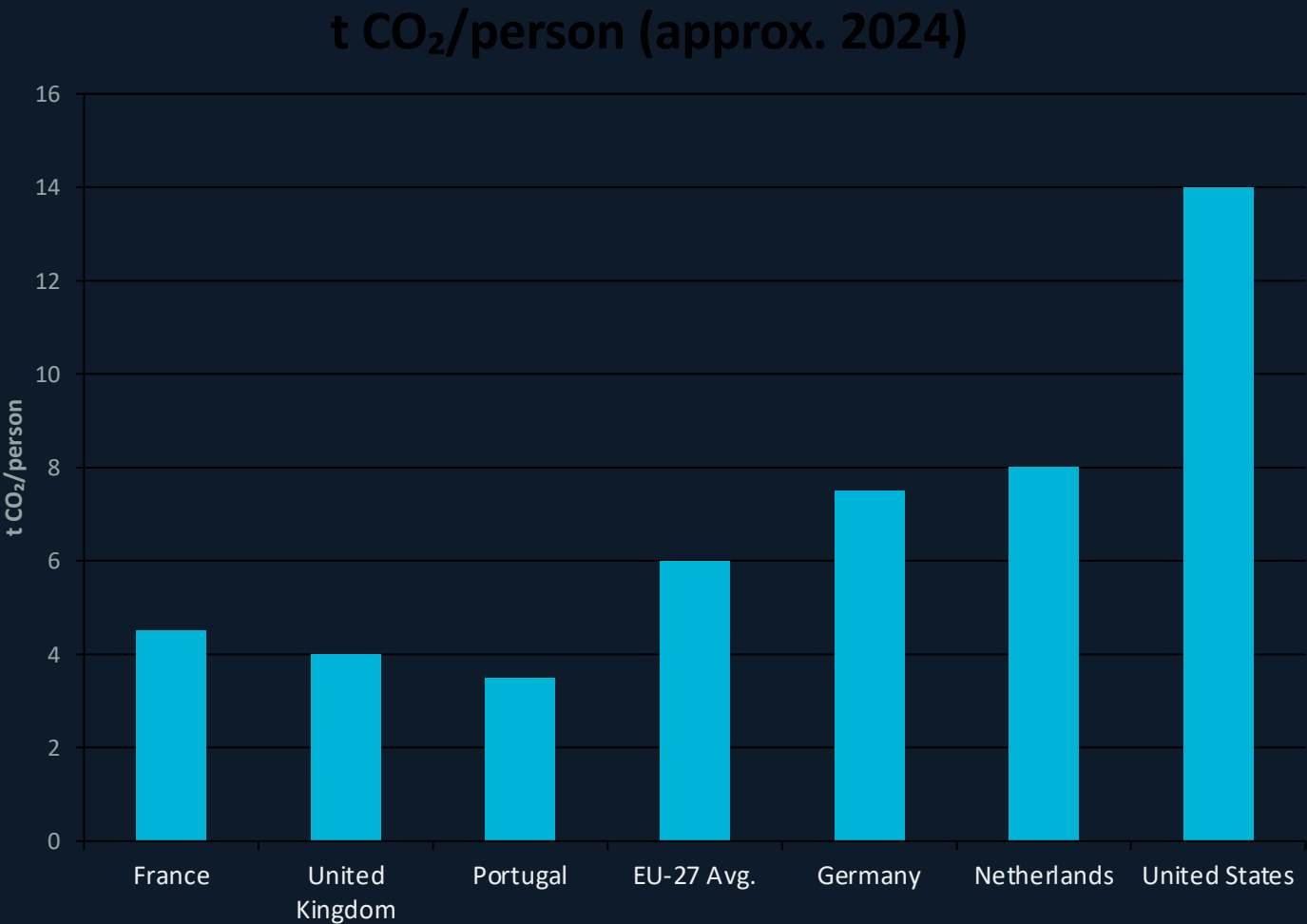
Key caveat

Gains in developed-nation reductions are partly offset by growth in Asia, Middle East and Africa.

Source: figure_5.jpg (Our World in Data) — choropleth map of 2024 annual change.

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Per capita CO₂ & energy structure, selected wealthy nations | Global Carbon Budget 2025



Why the gap?

- France**
~70% nuclear electricity → low grid emissions
- UK**
Rapid expansion of offshore wind + phase-out of coal power
- Portugal**
High share of hydro & wind power
- Germany**
Slower coal exit; higher industrial load
- Netherlands**
Gas-heavy electricity mix
- United States**
High per-capita consumption; fossil fuel reliance in power & transport

Key insight: Wealthy nations can decouple prosperity from emissions through nuclear and renewable energy — not just efficiency gains.

Key Takeaways: Responsibility Is Shared but Unequal

Global CO₂ Emissions: Who Emits, How Much, and What's Changing | 2024

1

Emissions Exceed 35 Bt/yr — No Peak Yet

Global fossil-fuel CO₂ surpassed 35 billion tonnes annually; growth has slowed but has not reversed.

2

China Is the Largest Emitter — But Not Per Capita

China emits ~31% of global CO₂ yet its per capita footprint (~9 t) remains well below the US (~14 t).

3

Historical Responsibility Dominated by Europe & the US

Europe and the US produced >90% of cumulative CO₂ in 1900; their historical share far exceeds today's ~29%.

4

150× Per Capita Gap Persists

From Chad / Niger (~0.1 t) to the US / Canada (~14 t): the least-responsible nations face the greatest climate risks.

5

Energy Policy Can Decouple Prosperity from Emissions

France & the UK demonstrate that high living standards do not require high emissions when low-carbon grids are prioritised.